



Wrocław University of Technology

IT Applications: Electronic media in business and commerce

dr inż. Dariusz Caban

dr inż. Tomasz Walkowiak

dr inż. Marek Woda



1. Introduction

- Course structure
 - lecture 2h/week
 - design project 1h/week
- Written assessment
 - Pre-session exam in week 15 (compulsory)
- Final score
 - 40% written exam
 - 60% design project marks
- Failure in the design project - disqualifies from exam
- Failed test can be taken again during the session
- Materials: e-portal



Subject educational outcomes

- | | |
|---------|---|
| PEK_W01 | knows the problems of e-business |
| PEK_W02 | knows the web-based technologies used in electronic commerce |
| PEK_W03 | knows the principles of operation of large information processing systems operating in the public sector and in supporting the commerce |
| PEK_W04 | knows the basic regulations of information security and the cryptographic tools required to ensure security |

Subject educational outcomes

- PEK_U01 is able to make the specification of a complex information system
- PEK_U02 can design the software for a commercial project, encompassing the security requirements
- PEK_U03 can implement an application for a commercial undertaking that makes use of the up-to-date web technologies and assess its security

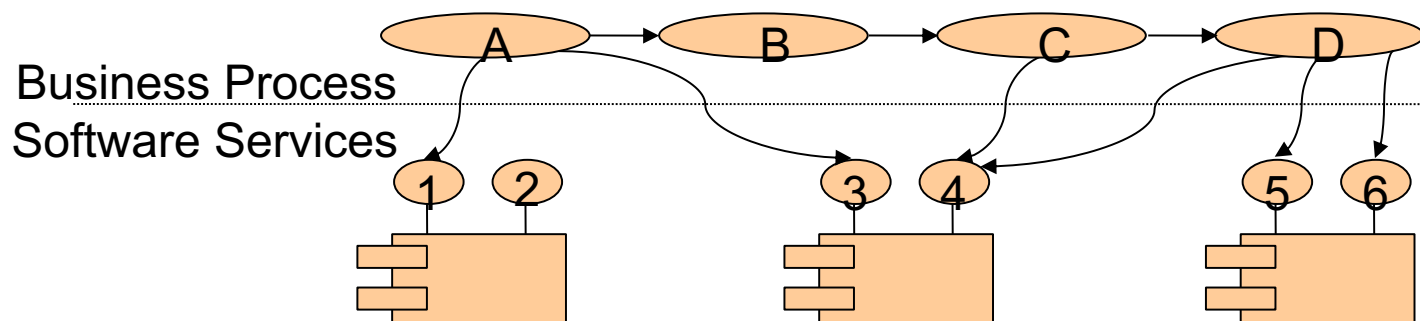


Subject educational outcomes

- | | |
|---------|---|
| PEK_K01 | is aware of the importance of the influence of modern technologies on the economic and social processes, and can critically analyze the related phenomena |
| PEK_K02 | can cooperate in a programming team, developing a complex information system, fulfilling various functions |
| PEK_K03 | can develop the schedule of programming tasks, assess their priorities and workloads, manage the risks of the project |
| PEK_K04 | understands the security risks connected with the application of IT |

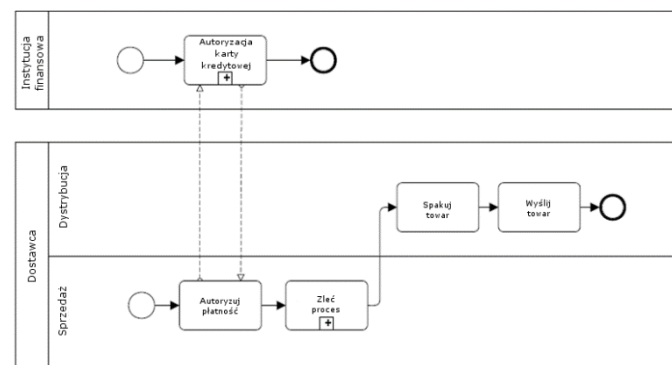
1. E-business and enterprise applications

- The architecture of enterprise applications
 - application servers
 - distributed programming (CORBA)



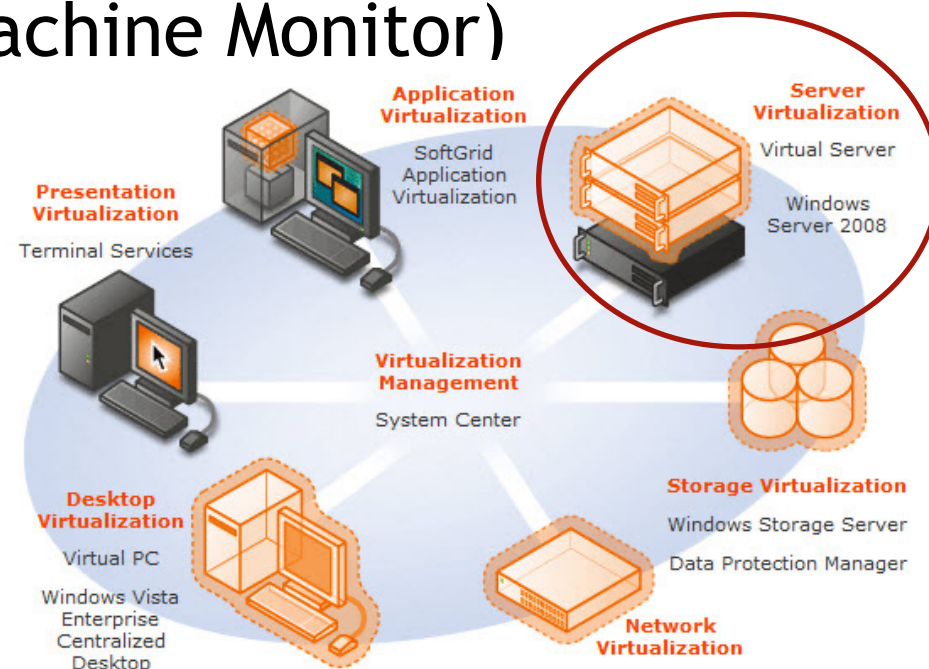
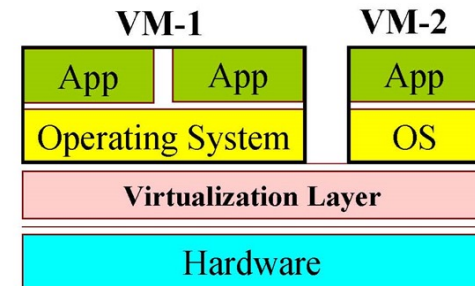
2. REST

- SOA
- REST architecture
- Implementation in Python
- Mirco-services
- GraphQL



3. Virtualization+CC

- History, background
- Types of virtualization
 - Full (binary translation), para-virtualization
 - Hardware Assisted Virtualization
- Hypervisor (Virtual Machine Monitor)
- Examples
 - VMware, Xen
- Security
- Cloud computing



4. Containers and GPU

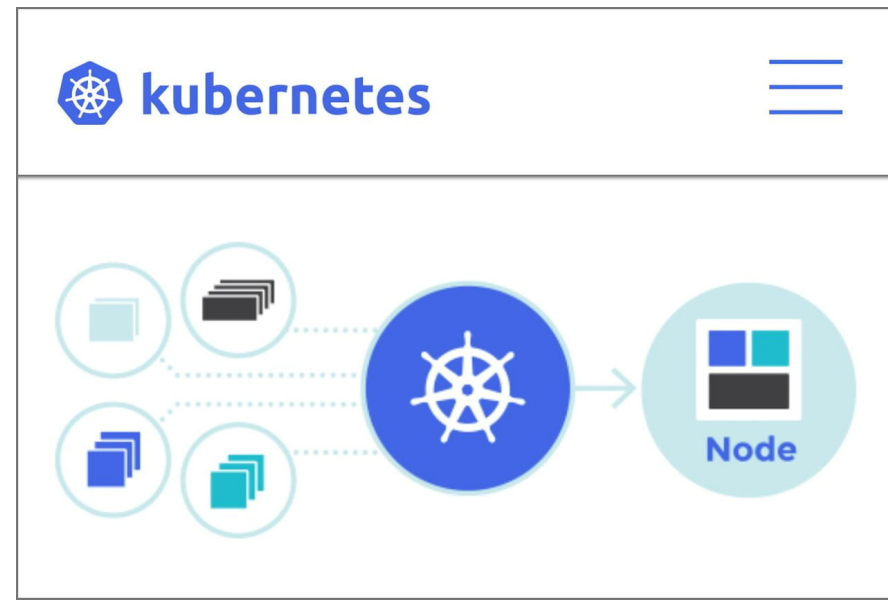
- Docker
- Docker-compose
- GPU in dockers
 - access to GPU
 - inference of deep models
 - ONNX



5. Orchestration

- Kubernetes
- Rancher
- CI

Tomasz Walkowiak





6. Basic software security mechanisms

- Cryptosystems, digests, session keys, MAC and HMAC control sums
- Digital signatures
- Message enveloping
- Certificates, trust models
- AAA mechanisms



7. Secure network protocols

- Layered model
- Secure networks requirements
- SSL protocol and traffic tunnelling
- HTTPS protocol
- Apache configuration supporting HTTPS



8. Secure electronic payments

- Payment card products, security of card payments
- Protocols supporting electronic payments (SET, 3D Secure)
- Vulnerabilities of WiFi networks



9. Economic aspects of securing systems

- Service vulnerabilities
- Technological and social aspects of security
- Security policy
- Assessment of the risk
- Investment into security



10. IT services management in the enterprise (MW)

1. ITIL - what is and what's not?
2. Essential Processes
 - *Incident Management*
 - *Problem Management*
 - *Change Management*
 - *Service Asset & Configuration Management*
 - *Knowledge Management*
 - *Release & Deployment*
 - *Event Management*
 - *Service Level Management*
3. KPI, Metcris

11. SLM, Availability and IT Services Continuity Management (MW)

1. Service Level Management

- Aim and application of SLM
- Scope of SLM
- Value for business
- Activities related to the process
- Types of contracts (OLA, UN, SLA)

2. Availability Management

- Understanding the essence of Availability Management
- Definitions: *Availability / Reliability / Maintainability / Serviceability*
- Calculation of the level of service availability
- Availability in “nines”
- Accessibility management, and other ITIL processes
- Cost of unavailability of services



12. SLM, Availability and IT Services Continuity Management (MW)

3. *Capacity Management* and *Service Continuity Management*

- Introduction to IT Service Continuity Management (ITSCM)
- Restoring services to operation (Cold, Warm, Hot reserves)

4. *Security vs. Safety*



13. *Blockchain and Cryptocurrencies*

1. **Blockchain essentials**

- *Block structure, Hash, Merkle Trees, Ledger*
- Distributed / decentralized systems

2. **History**

3. **Blockchain - why to use?**

4. **Cryptocurrencies**

- Bitcoin, Ethereum, Ripple & other
- History of payments systems
- BTC vs. fiat money

5. **Process of mining**

6. **Law and regulations**

7. **Cryptocurrencies Wallets**



14. Blockchain and Cryptocurrencies

- 1. BTC scaling issues**
- 2. ICO & DAO**
- 3. Consensus mechanisms**
- 4. Smart contracts**
- 5. Blockchain applications**